All-Event Dynamics of Finite Sticky Particles: Weighted Isotonic Geometry, Dissipation, and Weak Closure

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Abstract

We close the forward dynamics of an arbitrary finite one-dimensional system of positive point masses undergoing perfectly inelastic sticking. Initially coincident labels are canonically premerged, after which a single construction handles binary, multi-cluster, and spatially disjoint simultaneous events. Equivalently, positions are slopes of a lower convex hull in cumulative-mass coordinates. Every event conserves mass and momentum and has an exact pairwise-variance kinetic-energy loss.

1 Model, conventions, and source ownership

Let raw labels carry masses $m_i > 0$, ordered positions $x_1 \leq \cdots \leq x_N$, and velocities $v_i \in \mathbb{R}$. All labels at the same initial position are replaced at time zero by one canonical particle whose mass and velocity are

$$M = \sum_i m_i, \qquad V = M^{-1} \sum_i m_i v_i. \tag{1}$$

The resulting canonical positions are strictly increasing. This convention preserves mass and momentum and regards any raw-label energy drop as part of initialization. Zero mass is excluded because both (1) and the weighted geometry would otherwise have undefined blocks.

Canonical particles move ballistically until contact. Every maximal consecutive block occupying one position then becomes one particle, with mass and velocity given by (1); it never splits. Velocities at event times are right-continuous. This is a forward entropy dynamics: we make no backward-uniqueness claim.

The monotone-cone formulation and sticky semigroup are classical; a direct owner is Natile and Savaré [1]. Brenier, Gangbo, Savaré, and Westdickenberg treat sticky Lagrangian pressureless flows with interactions [2], and Hynd gives a Lagrangian pressureless-Euler construction [3]. The result below is a convention-complete, independently audited workspace certificate, not a claim of literature priority.

2 The all-event theorem

Write the canonical data again as $(m_i, x_i, v_i)_{i=1}^n$, now with $x_1 < \cdots < x_n$. Give $\mathbb{R}^n$ the weighted norm $\|z\|_m^2 = \sum_i m_i z_i^2$ and set

$$K = \{z \in \mathbb{R}^n : z_1 \leq \cdots \leq z_n\}.$$

Let P_K^m denote the unique projection in this norm.

Theorem 1 (Arbitrary finite all-event flow)**.** *For every canonical finite datum there is exactly one global forward sticky flow. Its labelled positions are*

$$X(t) = P_K^m(x + tv), \qquad t \geq 0. \tag{2}$$

The equal-coordinate blocks of $X(t)$ are precisely the physical clusters; their velocity is the mass-weighted average of the v_i in that block. The partitions only coarsen. Thus there are at most $n-1$ nontrivial mergers, while any one event time may contain several disjoint contact positions and any contact position may contain more than two incoming clusters.

Proof. The finite event construction is explicit. At any state, compare adjacent clusters A, B. If $V_A > V_B$, their prospective delay is

$$\tau_{A,B} = \frac{X_B - X_A}{V_A - V_B};$$

otherwise it is infinite. Advance every cluster through the least finite delay. At that time merge *all* maximal consecutive collocated blocks, including blocks at different positions. Each nontrivial merge lowers the cluster count, so the process is global after at most $n-1$ mergers. This also covers several adjacent prospective delays becoming equal.

It remains to identify the construction and prove uniqueness. Weighted pool-adjacent-violators starts from the free values $y_i(t) = x_i + tv_i$ and replaces every adjacent inversion by its mass-weighted mean. Its terminal blocks satisfy the variational inequalities for projection onto K, hence give the unique vector (2). Before the first contact no pool is needed. At first contact, exactly the maximal collocated incoming blocks are pooled, and their slope in t is their barycentric velocity. Repeating this argument after the event identifies the projection with the event construction on the next interval. Induction closes every interval. Once a set of indices is pooled, the same barycentric rule governs it in the next interval, so a physical block cannot split. The projection therefore selects the unique forward sticky continuation. $\square$

Separated clusters with equal velocity have infinite prospective delay and do not collide. A collision may fail to occur altogether. Simultaneity is not resolved by arbitrary pair ordering: maximal contact blocks are merged in one operation.

3 Lower convex hull and event geometry

Put $M_0 = S_0 = 0$ and, for $1 \le j \le n$,

$$M_j = \sum_{i \le j} m_i, \qquad S_j(t) = \sum_{i \le j} m_i(x_i + tv_i). \tag{3}$$

Take the greatest convex minorant of the polygonal line through $(M_j, S_j(t))_{j=0}^n$.

Proposition 1 (Hull representation)**.** *On each interval $[M_{a-1}, M_b]$ where the minorant has one affine face, its slope equals $X_i(t)$ for every $a \le i \le b$. Its faces are therefore exactly the sticky clusters at time t.*

Proof. The slope of the chord from M_{a-1} to M_b is the weighted mean of $y_a(t), \dots, y_b(t)$. Convexity orders successive slopes. Maximal faces therefore give a feasible isotonic vector. The minorant inequalities are the prefix-sum form of the projection variational inequality; equality at face endpoints supplies complementary slackness. Strict convexity of the weighted square norm identifies this vector with (2). $\square$

The hull picture also makes simultaneous events unambiguous. At an event, one or several neighboring groups of faces become collinear. Replacing each group by its maximal common face produces the unique post-event partition. Because later physical partitions coarsen, an equality face is never split by a label-level tie convention.

4 Exact conservation and dissipation

Suppose one event group contains incoming clusters with masses $M_a > 0$ and velocities V_a, $1 \le a \le r$. Set $M = \sum_a M_a$, $P = \sum_a M_a V_a$, and $\bar{V} = P/M$.

Proposition 2 (Multi-cluster loss identity)**.** *Mass and momentum are conserved, and the kinetic-energy loss is*

$$\Delta E = \frac{1}{2}\sum_a M_a V_a^2 - \frac{1}{2} M \bar{V}^2 = \frac{1}{2M} \sum_{a<b} M_a M_b (V_a - V_b)^2 \ge 0. \tag{4}$$

The formula adds over disjoint contact positions at the same event time. Globally, total mass and momentum are constant, and

$$\sum_C M_C X_C(t) = \sum_i m_i x_i + t \sum_i m_i v_i. \tag{5}$$

Proof. The definitions of M, P, and $\bar{V}$ prove mass and momentum conservation. Expanding $\sum_a M_a(V_a - \bar{V})^2$ gives the first difference in (4); symmetrizing gives the pairwise expression. Each term is nonnegative. Ballistic differentiation proves (5) between events, while event continuity and momentum conservation remove every jump. Disjoint event groups have disjoint incoming index sets, so their losses add. $\square$

This exact identity includes binary and arbitrary simultaneous mergers. It also distinguishes the physical entropy selection from a merely mass-conserving coalescence rule.

References

[1] L. Natile and G. Savaré, "A Wasserstein approach to the one-dimensional sticky particle system," *SIAM J. Math. Anal.* 41 (2009), 1340–1365, arXiv:0902.4373.

[2] Y. Brenier, W. Gangbo, G. Savaré, and M. Westdickenberg, "Sticky particle dynamics with interactions," *J. Math. Pures Appl.* 99 (2013), 577–617, arXiv:1201.2350.

[3] R. Hynd, "Lagrangian Coordinates for the Sticky Particle System," *SIAM J. Math. Anal.* 51 (2019), 3769–3795, doi:10.1137/19M1241775.